

```
my first inserted string'");
    ResultSet rs = stmt.executeQuery("select * from
mytable");
    // Retrieve and output the table data
    while (rs.next()) {
        System.out.println(rs.getString("mycolumn"));
    }
    conn.close();
} catch (Exception e) {
    e.printStackTrace();
}
}
```

This program can be compiled and executed, with the following two commands:

```
javac TestCUBRID.java
java TestCUBRID
```

It can be easily inferred from the code that connecting and handling CUBRID is very similar to other databases. So, if you have programmed for other databases in Java, you don't need to learn anything extraordinary in order to work with CUBRID.

An example in Python

To manipulate the CUBRID database with Python, you need CUBRIDdb, which is the Python extension package. The CUBRID Python driver can be installed in various ways. The easiest of them is to go with *easy_install*. The simple command for performing this task is shown below:

```
easy_install CUBRID-Python
```

A complete example with Python is shown below:

```
import CUBRIDdb
conn = CUBRIDdb.connect('CUBRID:localhost:33000:demodb',
'public', '')
cur = conn.cursor()

# Plain insert statement
cur.execute("INSERT INTO posts (id, title, body, last_
updated) VALUES (1, 'Title 1', 'Test body #1', CURRENT_
TIMESTAMP)")

# Parameterized insert statement
args = (2, 'Title 2', 'Test body #2')
cur.execute("INSERT INTO posts (id, title, body, last_
updated) VALUES (?, ?, ?, CURRENT_TIMESTAMP)", args)

conn.commit()
```

The complete information about handling CUBRID with Python is available at <https://www.python.org/dev/peps/pep-0249/>.

GUI tools for CUBRID

There are various graphical user interface tools available for handling CUBRID in an easier manner. These are listed in Table 1.

Table 1

Tool	Description
CUBRID Manager	This is a database administration tool developed for CUBRID. It is available with a BSD licence.
CUBRID Query Browser	This is a scaled-down version of CUBRID Manager.
CUBRID Web Manager	This is a Web based version of CUBRID Manager. It can be deployed on remote servers.
CUBRID migration toolkit	Data migration is an important task when handling multiple databases. The CUBRID migration toolkit enables the migration of data from other databases such as MySQL, or from the previous versions of the CUBRID database to the latest versions.

CUBRID HA

What 'HA' refers to is 'high availability'. This is a feature that enables uninterrupted service in scenarios when there is failure in either the hardware, software or the network. CUBRID HA includes more than one server system. These are used to provide uninterrupted services when an individual server fails. CUBRID HA synchronises databases among multiple servers. This feature can be enabled for mission-critical applications when there is a need for 24/7 availability of service. More information on CUBRID HA is available at <https://www.cubrid.org/manual/en/10.1/ha.html>.

If you want to contribute to CUBRID RDBMS, then check out the source from <https://github.com/CUBRID>. Based on your expertise, you can get involved in contributing to the development of drivers, tools, content management systems (CMS), etc. Even as just a user, you can report bugs by opening issues in JIRA or by simply sending an email to bug@cubrid.org.

CUBRID is an actively maintained relational database engine. Recent updates on it are available at <https://www.cubrid.org/blog>. 

References

- [1] <https://www.cubrid.org/>
- [2] <https://www.cubrid.org/documentation>

By: Dr K.S.Kuppusamy

The author is an assistant professor of computer science at Pondicherry Central University. He has 13+ years of teaching and research experience in academia and industry. He can be reached at skkuppup@gmail.com.